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期末课程评教入口二维码



EE115B *Digital Circuits*

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Sequential Logic

Basic SR Latch

Gated SR Latch

Limitations of Combinational Logic

Can we build a circuit using combinational logic?

When a button is pushed:

Turn on the light when it is off

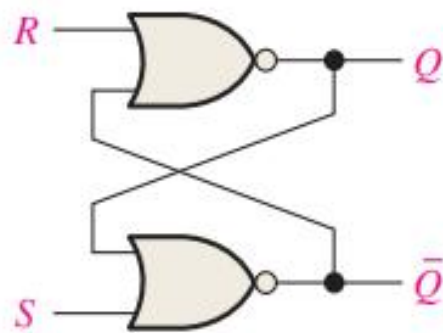
Turn off the light when it is on.

Sequential logic circuits

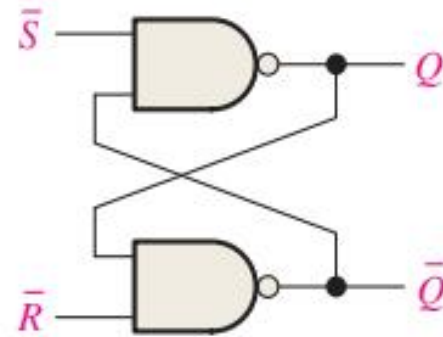
Have memory and their outputs depend on both current and past inputs, allowing for the storage and manipulation of information over time.

Basic SR Latch

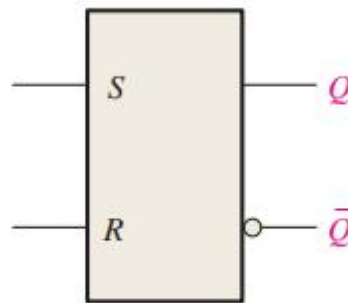
SR Latch: a digital logic circuit that has two stable states and can store one bit of data



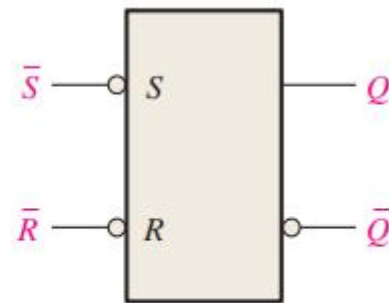
(a) Active-HIGH input S-R latch



(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch



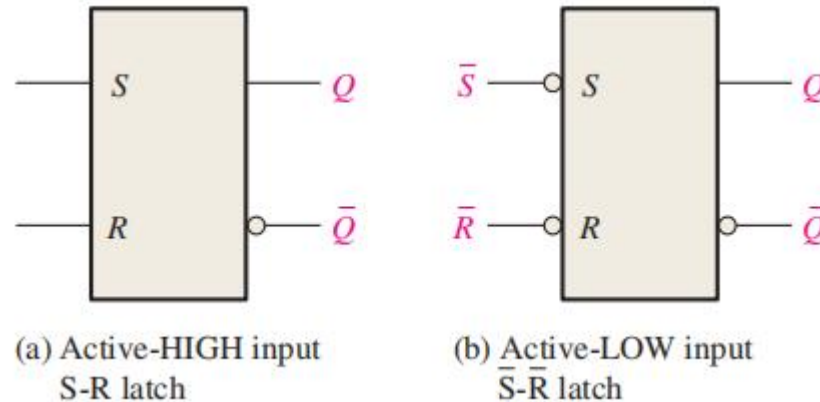
(a) Active-HIGH input S-R latch



(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

Basic SR Latch: Its Function

Function of SR Latch



Characteristic Table: Describes next state given input and current state. (Similar to truth table, but focus on states)

TABLE 7-1

Truth table for an active-LOW input $\bar{S}$ - $\bar{R}$ latch.

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition

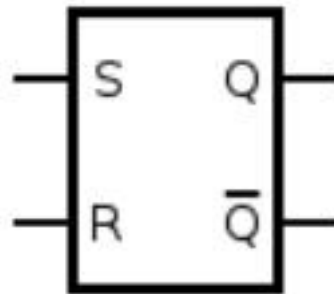
Basic SR Latch: Active Low vs Active High

TABLE 7-1

Truth table for an active-LOW input $\bar{S}$ - $\bar{R}$ latch.

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition

Active low



Active high

S	R	Q_a	Q_b	
0	0	0/1	1/0	(no change)
0	1	0	1	
1	0	1	0	
1	1	0	0	

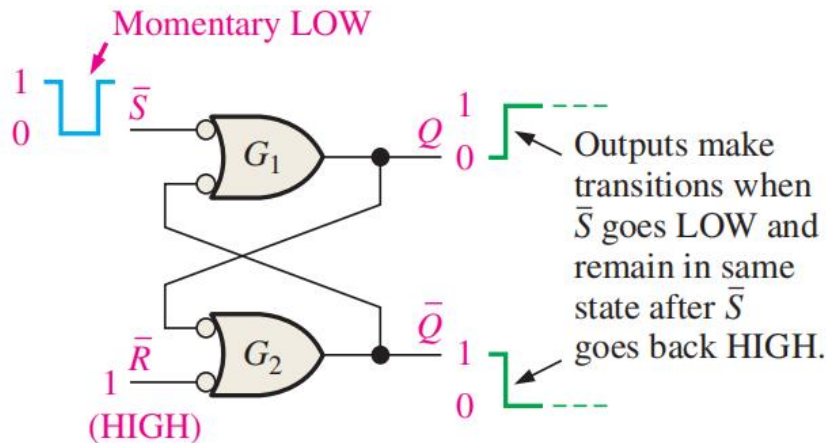


Basic SR Latch: How circuit works?

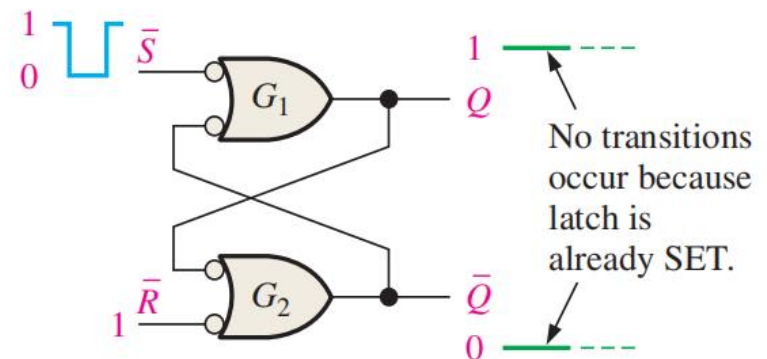
TABLE 7-1

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Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition



Latch starts out RESET ($Q = 0$).



Latch starts out SET ($Q = 1$).

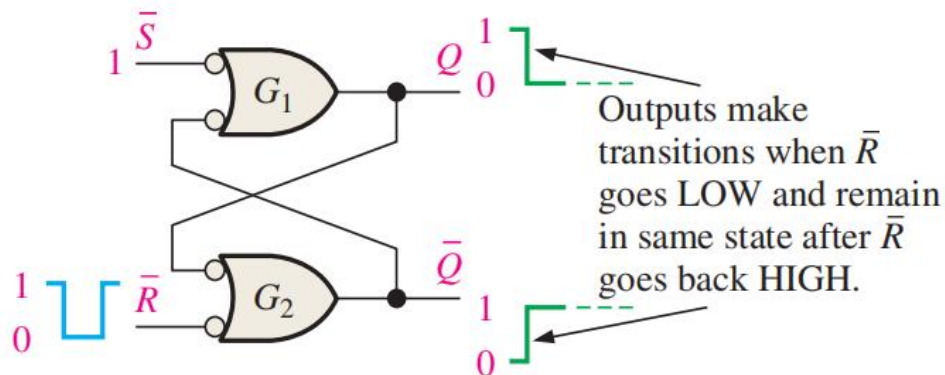
(a) Two possibilities for the SET operation

Basic SR Latch: How circuit works?

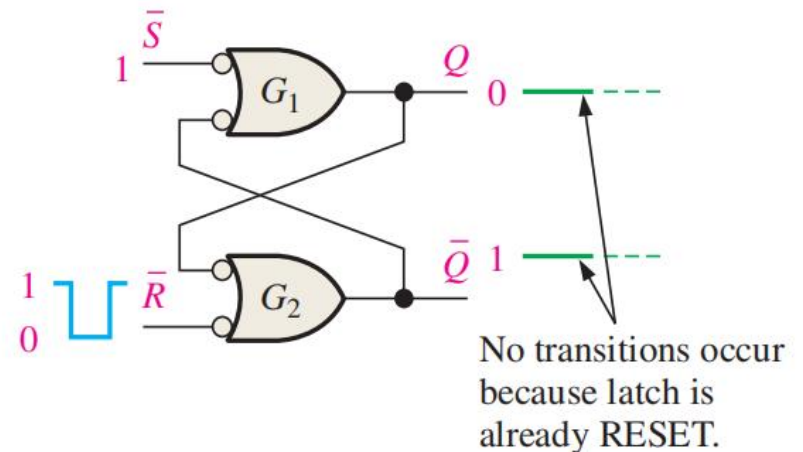
TABLE 7-1

Truth table for an active-LOW input $\bar{S}$ - $\bar{R}$ latch.

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition



Latch starts out SET ($Q = 1$).



Latch starts out RESET ($Q = 0$).

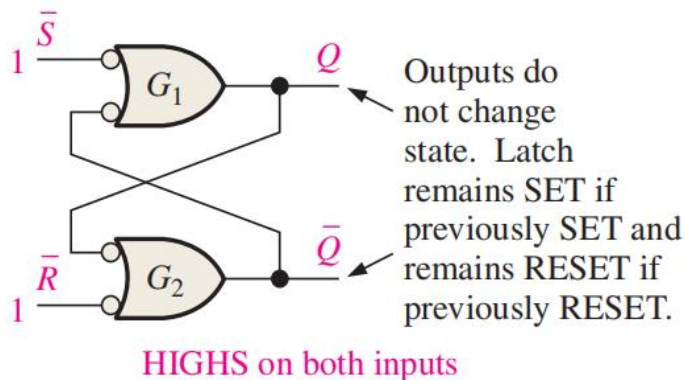
(b) Two possibilities for the RESET operation

Basic SR Latch: How circuit works?

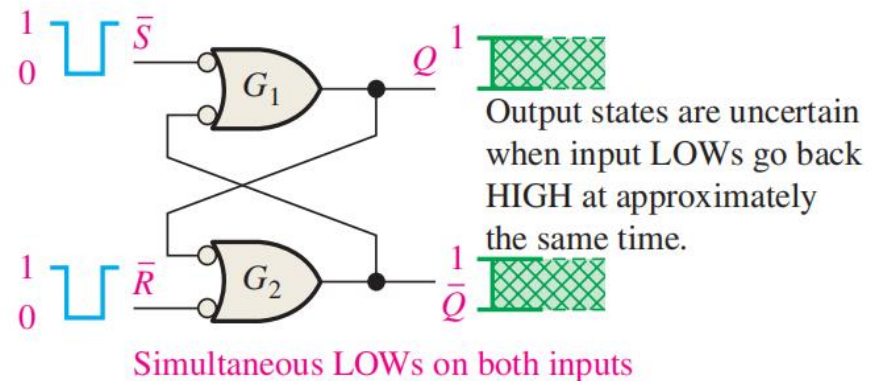
TABLE 7-1

Truth table for an active-LOW input $\bar{S}$ - $\bar{R}$ latch.

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition



(c) No-change condition



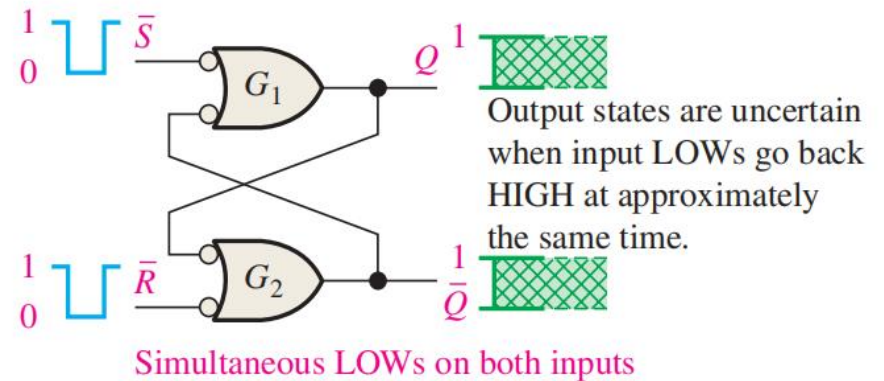
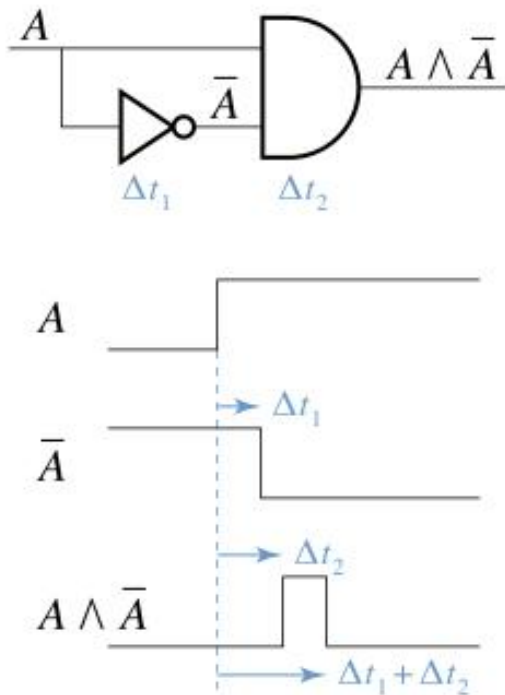
(d) Invalid condition



Race Hazard Conditions

- A hazard is an unwanted temporary change (glitch or spike) in the output of a digital circuit
- A race hazard occurs when two or more signals change simultaneously and compete to control the output. Because of unequal propagation delays, the final output depends on which signal arrives first, potentially causing unpredictable behavior.

Race Hazard: Examples



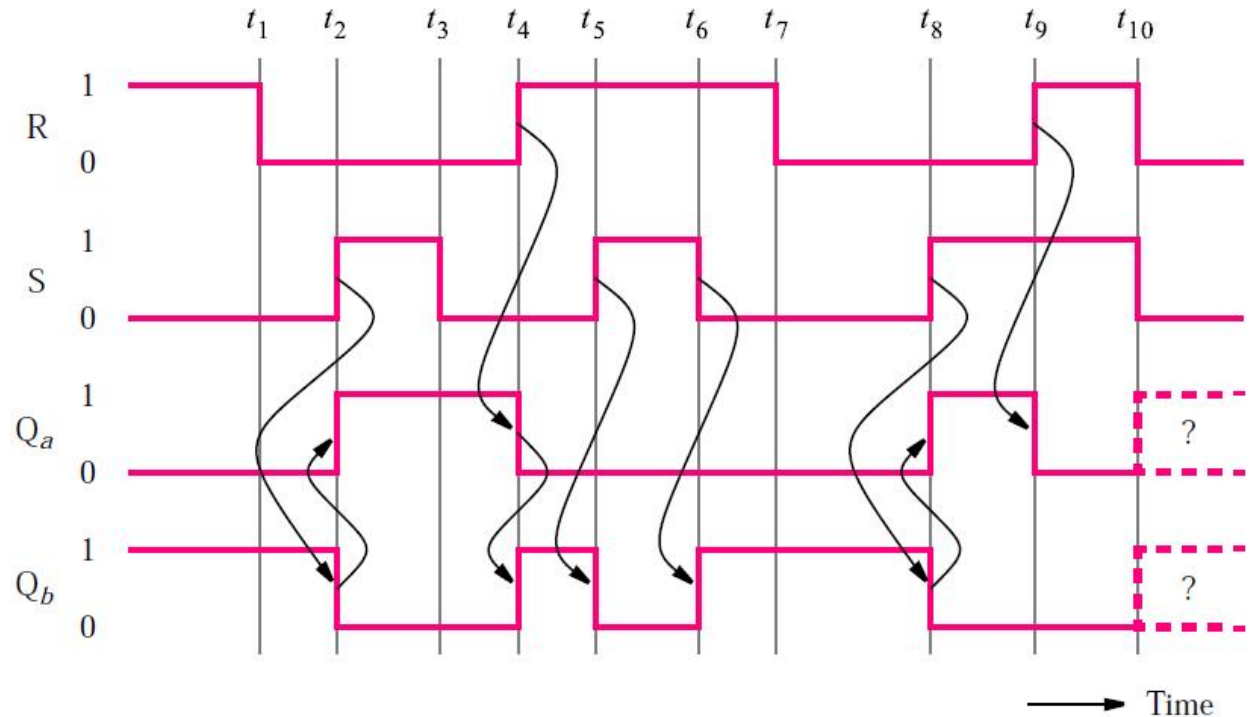
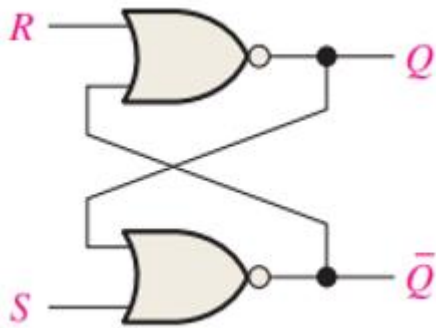
(d) Invalid condition

https://en.wikipedia.org/wiki/Race_condition



Basic SR Latch: Timing Diagram

Timing diagram (assume no propagation delay)



S	R	Q_a	Q_b	
0	0	0/1	1/0	(no change)
0	1	0	1	
1	0	1	0	
1	1	0	0	

Basic SR Latch: VHDL

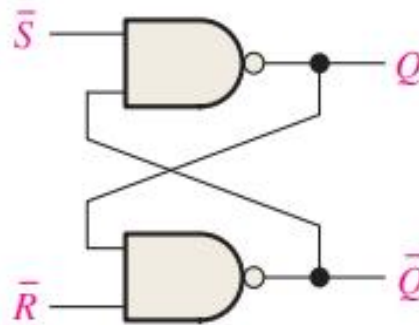
VHDL implementation



```
entity SRLatch is
  port (SNot, RNot: in std_logic; Q, QNot: inout std_logic);
end entity SRLatch;

architecture LogicOperation of SRLatch is
begin
  Q <= QNot nand SNot;
  QNot <= Q nand RNot;
end architecture LogicOperation;
```

SNot: SET complement
RNot: RESET
complement
Q: Latch output
QNot: Latch output
complement

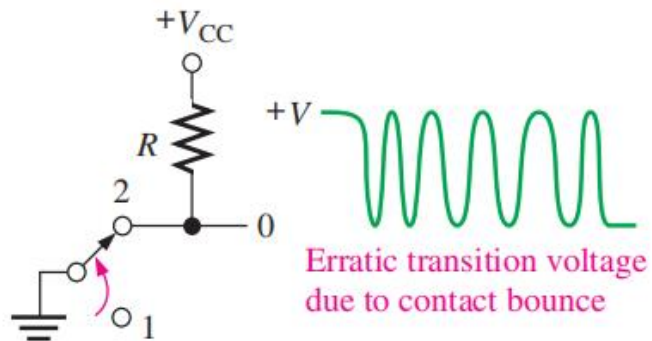


(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

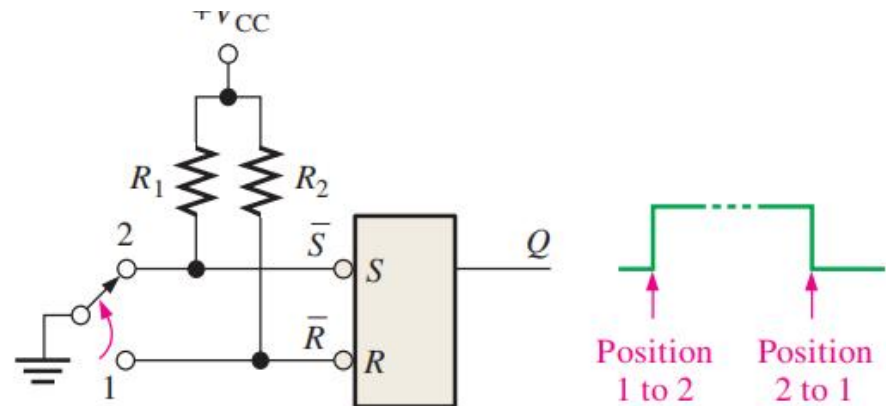


Basic SR Latch: Applications

Application: eliminate the effects of switch bounce



(a) Switch contact bounce

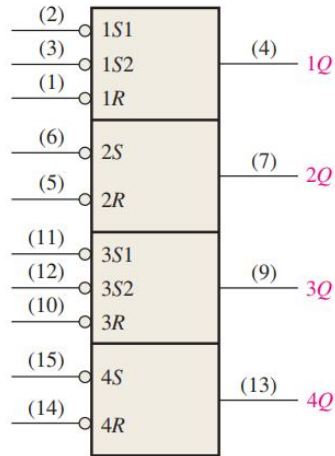


(b) Contact-bounce eliminator circuit

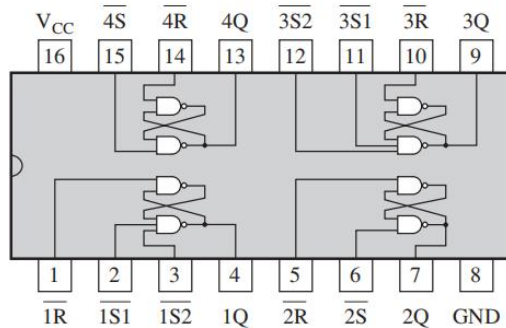
If bounce happens: 1 1 = no change



Basic SR Latch: Applications



(a) Logic diagram



(b) Pin diagram

74HC279A

Function Table

Inputs		Output
$\bar{S}^{*2}$	$\bar{R}$	Q
H	H	Q_0
L	H	H
H	L	L
L	L	H^{*1}

H : High level

L : Low level

Q_0 : The level of Q respectively, before the indicated steady-state input conditions were established.

Notes: 1. It is unpredictable, if $\bar{S}$ or $\bar{R}$ goes High.

2. As to latches which has two $\bar{S}$ inputs.

H: Both of $\bar{S}$ inputs are high.

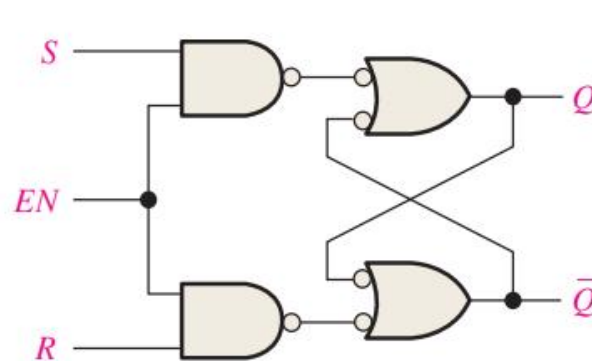
L: Either or both of $\bar{S}$ inputs are low.



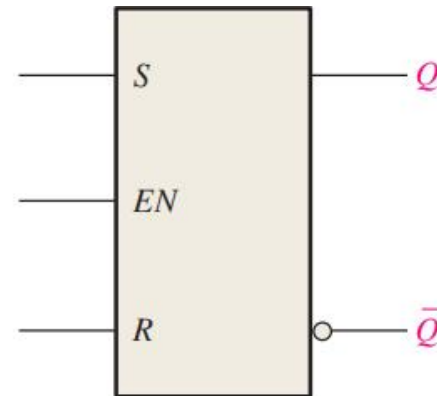
Gated SR Latch

Gated SR latch: Add a **Clock** (or named **Enable**) input
(avoid glitches, allowing better synchronization)

Example: bus sender and receiver synchronization.



(a) Logic diagram



(b) Logic symbol

Control signal name: EN or CLK

Clk=0, disabled: latched

Clk=1, enabled: basic SR latch

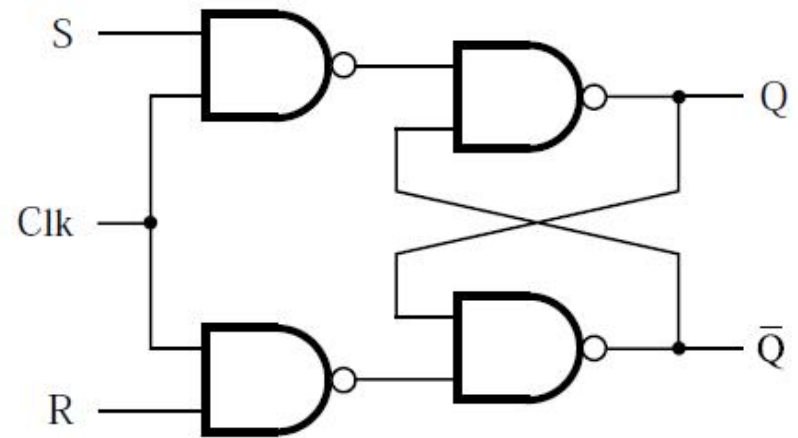
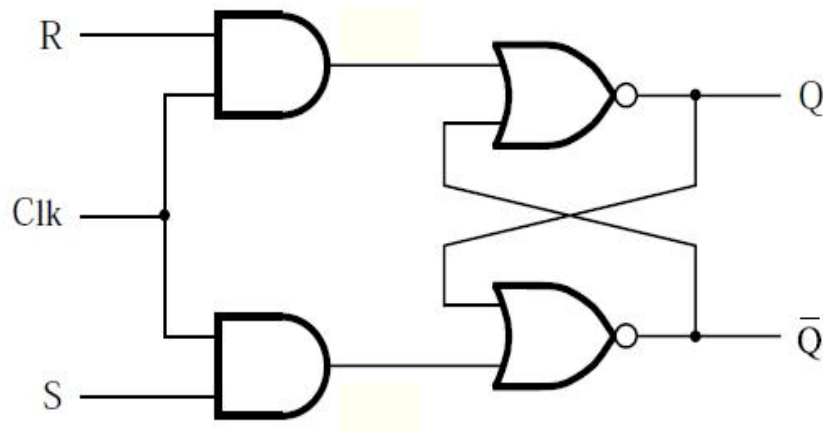
Clk	S	R	$Q(t+1)$
0	x	x	$Q(t)$ (no change)
1	0	0	$Q(t)$ (no change)
1	0	1	0
1	1	0	1
1	1	1	x



Gated SR Latch

Two implementations

- NOR gates
- NAND gates



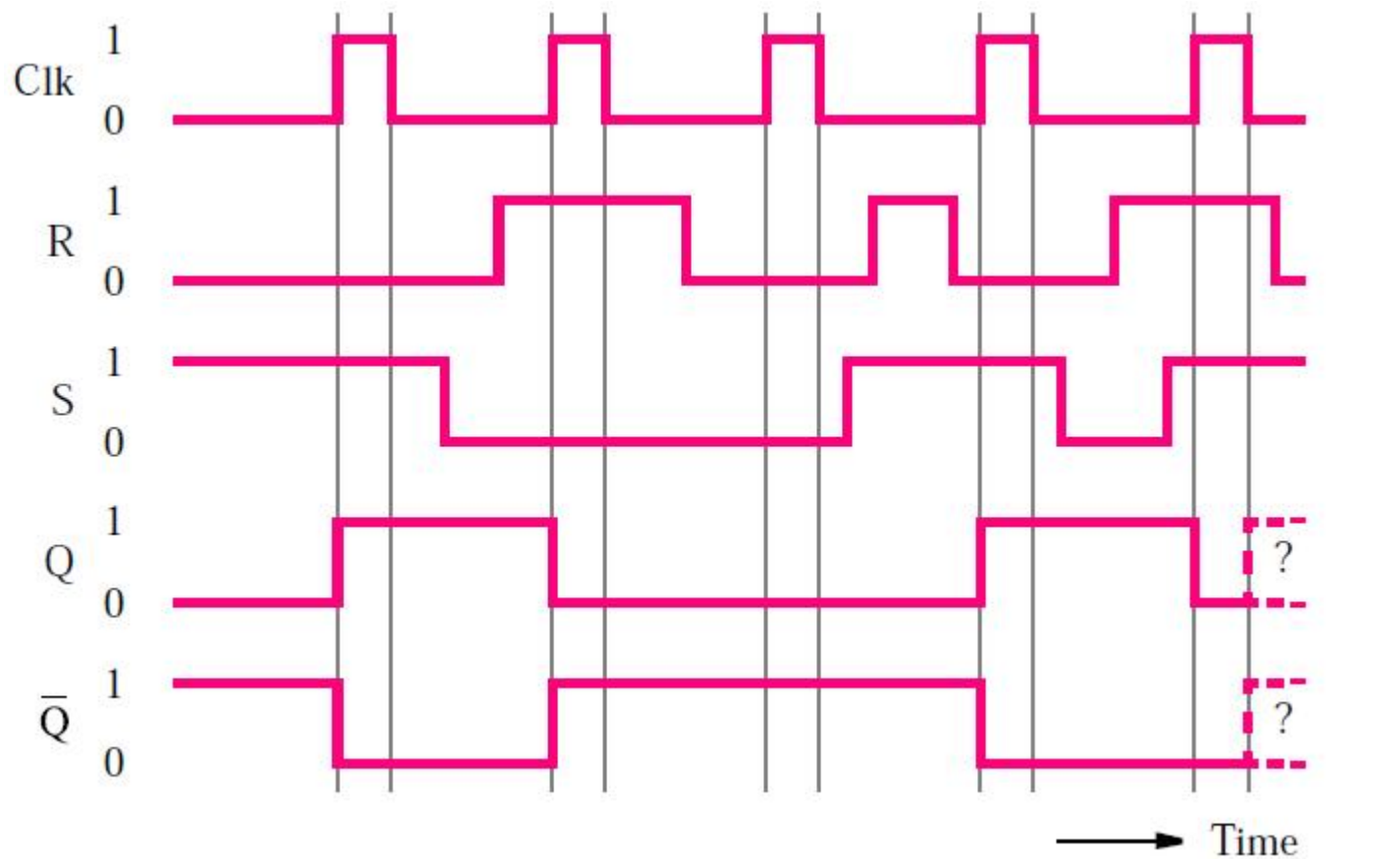
NAND gate implementation

Clk	S	R	Q^+	Comments
0	x	x	Q	No change, typically stable states $Q = 0, \bar{Q} = 1$ or $Q = 1, \bar{Q} = 0$
1	0	0	Q	No change, typically stable states $Q = 0, \bar{Q} = 1$ or $Q = 1, \bar{Q} = 0$
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	x	Avoid this setting



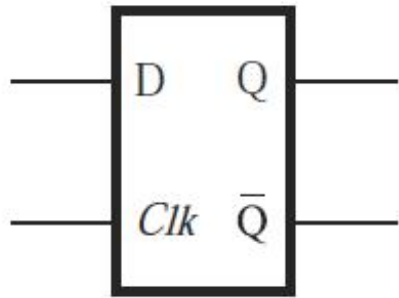
Gated SR Latch: Timing diagram

- Timing diagram
- Q only changes when CLK=1

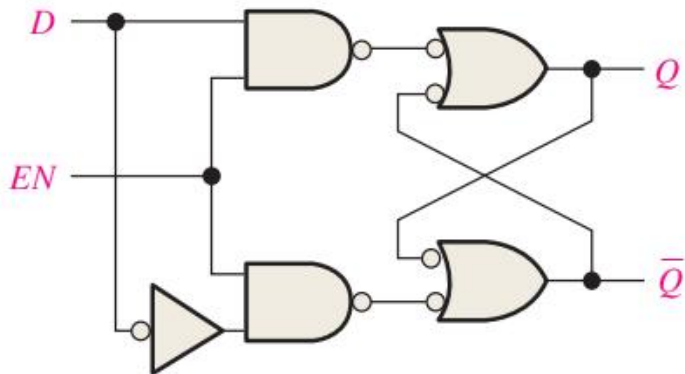


Gated D Latch

- Foundation for D Flip Flop
- Memory-like input (intuitive).
- Mitigate Race Hazard problem.



Clk	D	$Q(t+1)$
0	x	$Q(t)$
1	0	0
1	1	1

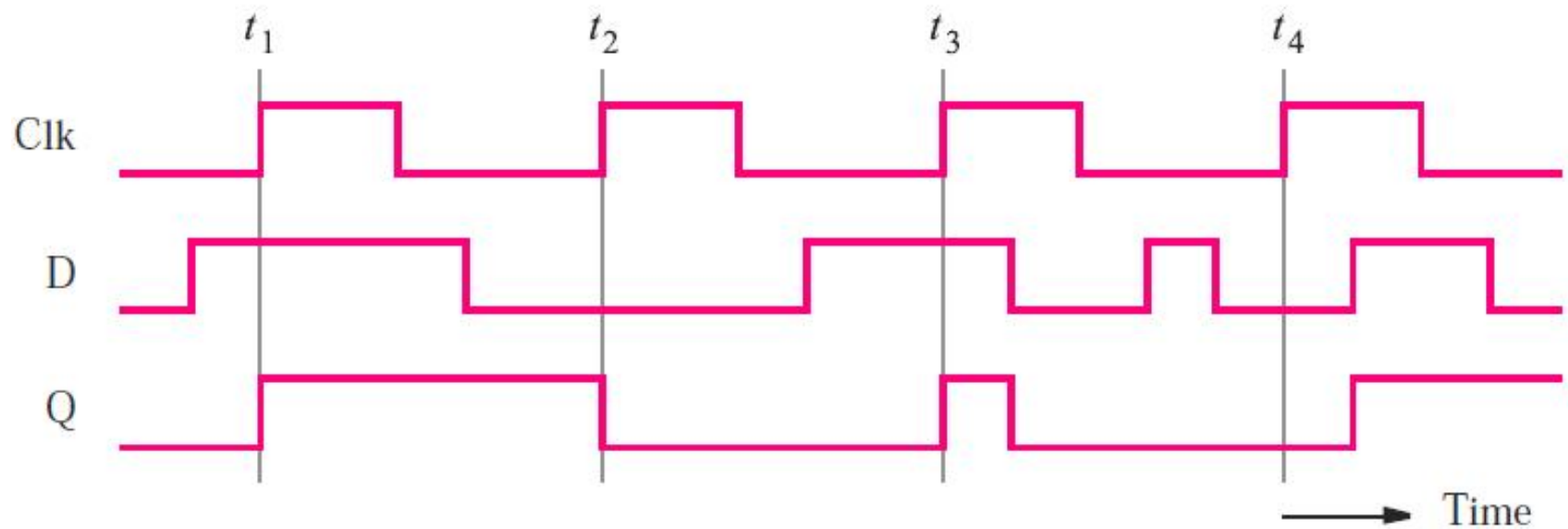


Only one input: D (data)

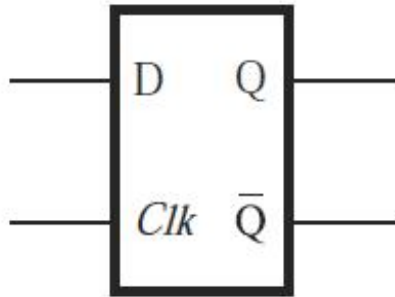
- Clk=1: output (Q) follows input (D)
- Clk=0: Q holds its previous value

Gated D Latch

Timing diagram



VHDL code for gated D Latch



Clk	D	Q(t + 1)
0	x	Q(t)
1	0	0
1	1	1

```
LIBRARY ieee ;
USE ieee.std_logic_1164.all ;

ENTITY latch IS
    PORT ( D, Clk : IN  STD_LOGIC ;
          Q      : OUT STD_LOGIC) ;
END latch ;
```

```
ARCHITECTURE Behavior OF latch IS
BEGIN
    PROCESS ( D, Clk )
    BEGIN
        IF Clk = '1' THEN
            Q <= D ;
        END IF ;
    END PROCESS ;
END Behavior ;
```

